

Clinicopathological Report

Paediatric uveal melanoma

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ABSTRACT

Uveal melanoma is extremely rare in the paediatric population and can be associated with various pre-existing conditions. We report the case of a 9-year-old girl with no predisposing factor who presented with choroidal melanoma. A review of the literature is presented and various clinical, histopathological and prognostic features of paediatric uveal melanoma are discussed.

Key words: chromosomal abnormality, paediatric malignancy, uveal melanoma.

INTRODUCTION

Uveal melanoma is the most common primary ocular malignancy in adults.¹ Even though there has been one published case of congenital uveal melanoma,² it is extremely rare in children under the age of ten. In a large series of 8000 patients with uveal melanoma,³ only 63 patients (0.8%) were aged 20 years or younger. Among these patients, only 10 patients (16% of children or 0.12% of all patients) were under the age of 10 years. In adults, the prognosis for uveal melanoma is known to be associated with several clinical, histopathological and cytogenetic factors.⁴ However, there are little data regarding the prognosis or prognostic features of uveal melanoma in children.

CASE REPORT

A 9-year-old girl presented with a 4-week history of reduced vision in her right eye. She was previously well with no significant past medical history. Her visual acuity in the right eye was counting fingers unaided and 6/36 with pinhole. Visual acuity in the left eye was 6/4.5. On slit-lamp examination, a large, smooth, dome shaped pigmented lesion arising from the superior peripheral choroid was noted (Fig. 1). Sentinel vessels were not prominent. There was associated serous retinal detachment which involved the macula and extended inferiorly.

An A/B-scan demonstrated a corresponding smooth, dome-shaped elevation of the retinal contour measuring approximately 17 mm in basal diameter and 13 mm in maximal thickness. Internal reflectivity was moderate to low with a positive angle kappa (Fig. 2). Doppler studies were also performed and no vasculature or internal circulation could be seen in the lesion. An MRI scan showed a very smooth outline of the lesion with a homogenous consistency. No vascularity could be identified. There was no evidence of any other primary/metastatic lesions on systemic evaluation.

After discussion with the patient's parents and the ocular oncology, oculoplastics and paediatric specialists a decision was made to proceed with fine needle biopsy. As this was inconclusive, enucleation of the right eye and insertion of a vicryl mesh covered Medpore implant was performed. Histopathology (Fig. 3) demonstrated a choroidal melanoma with spindle cell morphology and mild cellular

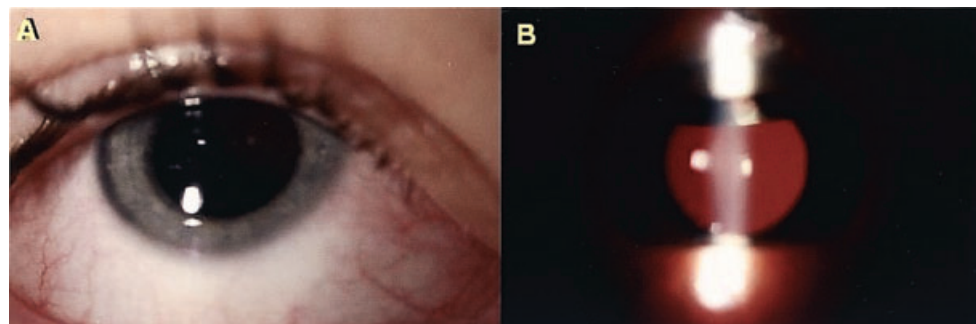


Figure 1. Slit-lamp appearance of the lesion.

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pleomorphism. Melanin pigmentation was uniform and heavy but, the cells lacked the large polygonal shape and bland nuclear morphology of melanocytoma. Mitoses were present and numbered up to 2 per 10 high power field and the proliferation index (Ki67/MIB-1) was 4–5%. The tumour involved the pars plana of the ciliary body but did not reach the anterior chamber angle. There was no extraocular extension. Fluorescent *in situ* hybridization revealed a signal pattern consistent with gain of the 8q locus. There was no evidence of loss of the chromosome 3 locus.

Since her surgery approximately 18 months ago, she has been regularly reviewed by her paediatric oncologist and ophthalmic oncologist and to date there is no evidence of local or systemic recurrence of disease.

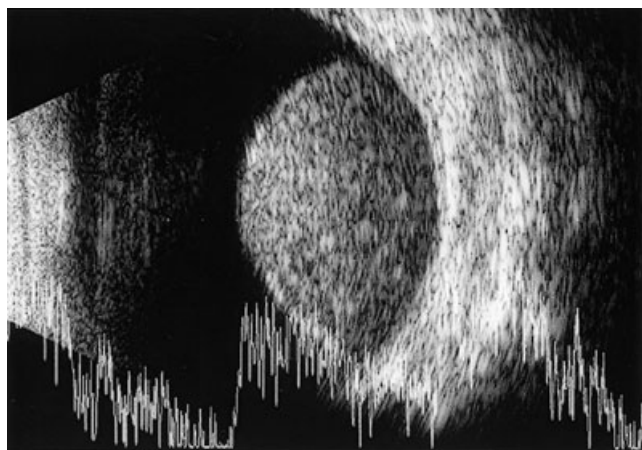
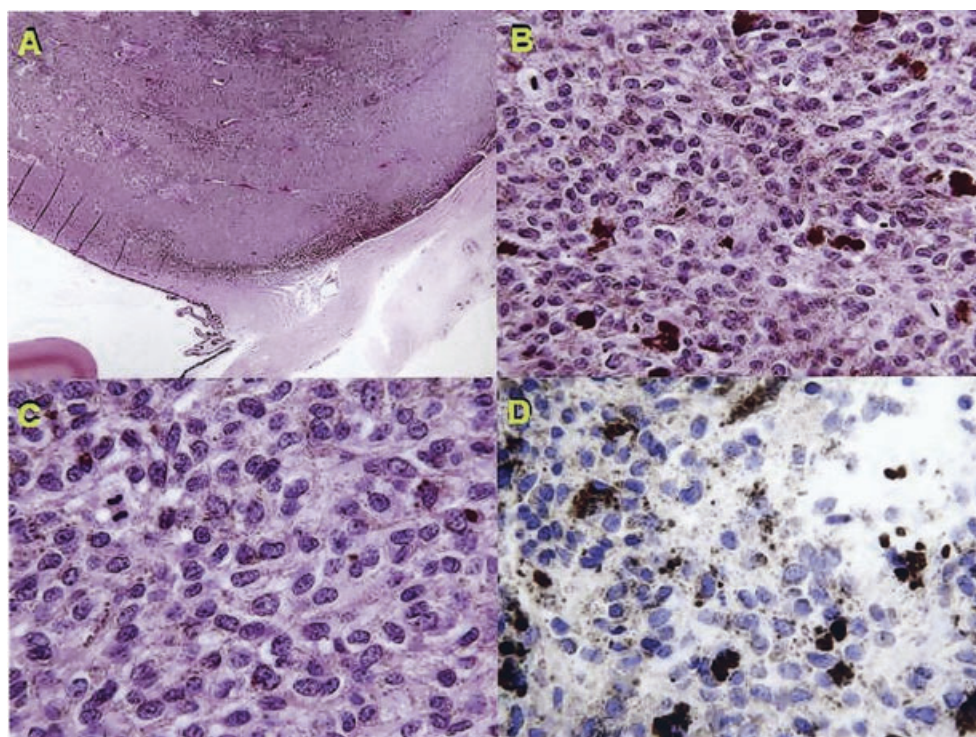


Figure 2. A/B-scan appearance of the lesion.

Figure 3. Histopathological appearance of the lesion. (a) Low magnification view of the lesion showing the choroidal tumour invading the ciliary body. (b) High magnification view showing the pigmented spindle cells with pleomorphism and scattered heavily pigmented melanophages. (c) Another high magnification view of the tumour cells showing a mitotic figure near the top left hand corner. (d) Ki67/MIB-1 stain. The proliferation index was reported as 4–5%.



COMMENT

Uveal melanoma in children is extremely rare. Only one case series³ and a few case reports^{5–13} have been published since 1980 reporting uveal melanoma in children (see Table 1). The clinical and histopathological features of uveal melanoma in children are similar to that in adults. It is usually unilateral and affects both eyes equally. No gender difference has been observed. It more commonly affects the choroid (54%) followed by the iris (25%).³ Spindle cell type may be more commonly found in uveal melanoma in children (46%).³ The differential diagnosis of paediatric uveal melanoma includes medulloepithelioma, melanocytoma, neuroepithelial cysts, haemangiomas and haematomas and acquired inflammatory masses. Conditions that have been described to be associated with the development of uveal melanoma include ocular melanocytosis, neurofibromatosis type I, familial melanoma or cutaneous dysplastic naevus syndrome.¹⁴ Significantly, in the case series³ reported by Singh *et al.*, oculo(dermal) melanocytosis was observed to be nine times more common in young patients with uveal melanoma than the general population with uveal melanoma. Our patient did not demonstrate any of these conditions.

There are very limited data on the long-term prognosis in these children. Non-random cytogenetic changes of uveal melanoma in young patients are poorly characterized and, to our knowledge, there are no other reported cases of chromosomal abnormalities in paediatric uveal melanoma in the literature. Monosomy in chromosome 3 and gain of chromosome 8 are markers of poor prognosis in adult uveal melanoma, and may imply a similarly poor prognosis in paediatric uveal melanoma. The cytogenetic profile of this case is dif-

Table 1. Reported cases of paediatric uveal melanoma 1980–2006

Source	Age/ gender	Site	Largest diameter /thickness	Presenting symptom	Histopathology	Treatment	Local spread	Follow up /progress	Associated condition
Soni <i>et al.</i> ¹⁰	8/F	Choroid	17 mm/10 mm	Reduced vision	Mixed cell type	Enucleation	Restricted to Bruch's membrane	Liver metastases in 1 year	Dysplastic naevus syndrome
Gundaz <i>et al.</i> ⁶	14/M	Choroid	10 mm/3 mm	Routine examination	Spindle cell type	Enucleation	Nil	NR	Ocular melanocytosis
Rosenbaum <i>et al.</i> ⁸	5/M	Iris	diffuse	Iris mass, reduced vision	Mixed cell type	Enucleation	Nil	Metastasis to nasopharynx and base of the skull in 8 months	Nil
Tran <i>et al.</i> ¹¹	12/M	Iris	3 mm/NR	Incidental finding	Mixed cell type	Enucleation	Iridocorneal angle, ciliary body and inner layer of the anterior sclera	NR	Nil
Shields <i>et al.</i> ⁹	16/M	Choroid	7 mm/3.6 mm	Routine examination	NR	Plaque radiotherapy	NR	NR	Immunosuppressed, minimal change nephrotic syndrome
Singh <i>et al.</i> ³	16/F	Choroid	13 mm/8 mm	Reduced vision	Spindle type	Enucleation	Nil	NR	Cutaneous melanoma
Demirci <i>et al.</i> ⁵	9/F	Ciliary body	NR/10 mm	Leucocoria and exotropia	Mixed cell type	Enucleation	Nil	Nil metastasis in 10-year follow up	Nil
Gunduz <i>et al.</i> ⁷	8/F	Choroid	18 mm/9.6 mm	Reduced vision	Mixed	Enucleation	Nil	Liver metastasis in 1 year	Ocular melanocytosis
Singh <i>et al.</i> ¹²	10/M	Iris	4.5 mm/NR	Routine examination	Mixed cell type	Local resection	Nil	NR	Familial atypical mole- melanoma (FAM-M)
Shields <i>et al.</i> ¹³	6/F	Choroid and ciliary body (2 lesions)	9.0 mm/4.1 mm and 6.0 mm/2.2 mm	Routine examination	Mixed cell type	Enucleation	Early invasion of a scleral canal without extrascleral invasion	NR	syndrome Mild sectorial scleral and uveal melanocytosis

F, female; IL-2, interleukin 2; M, male; NR, not reported.

difficult to interpret, however, along with the spindle morphology and low Ki67 proliferation index, may represent a less aggressive tumour biology.

Certain features of this case deserve attention. First, although rare, uveal melanoma in children is well recognised. Second, although this patient did not have any obvious predisposing factors, ocular (dermal) melanocytosis is much more common in the children with uveal melanoma compared with adults and children with oculodermal melanocytosis warrant long-term monitoring. Finally, there is limited knowledge regarding the long-term prognosis and significance of prognostic cytogenetic features in these children.

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